



The illustration depicts a spacious, modern home goods store. In the foreground, a man on the left examines a product on a shelf, holding a blue tote bag that says "Good things inside." To his right, a woman is looking at her phone while holding a large white tote bag with the text "Love Where You Live." and a brown paper shopping bag that says "Live Well. Shop Smart." The store is filled with wooden shelves stocked with various home products, including bottles, boxes, and plants. A central display table holds more products, including a potted plant and several boxes. In the background, other customers are browsing. Signs throughout the store include "Less clutter, more life." on the left, "Made for living." at the top center, "Kitchen & Dining", "For the way you live.", "Bath & Body", and "Storage & Organize" on the right. A central sign reads "Live Well. Shop Smart."

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

Dataset Size

Rows: 3,900

Columns: 18

Missing Data: 37 values in Review Rating column

Key Features

- **Customer demographics** (Age, Gender, Location, Subscription Status)
- **Purchase details** (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- **Shopping behavior** (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

Data Loading Imported the dataset using <code>pandas</code>.	Initial Exploration Used <code>df.info()</code> to check structure and <code>.describe()</code> for summary statistics.
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	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)
count	3900.000000	3900.000000	3900	3900	3900	3900.000000
unique	NaN	NaN	2	25	4	NaN
top	NaN	NaN	Male	Blouse	Clothing	NaN
freq	NaN	NaN	2652	171	1737	NaN
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392
min	1.000000	18.000000	NaN	NaN	NaN	20.000000
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000

	Location	Size	Color	Season	Review Rating	Subscription Status
count	3900	3900	3900	3900	3863.000000	3900
unique	50	4	25	4	NaN	6
top	Montana	M	Olive	Spring	NaN	No
freq	96	1755	177	999	NaN	2847
mean	NaN	NaN	NaN	NaN	3.750065	NaN
std	NaN	NaN	NaN	NaN	0.716983	NaN
min	NaN	NaN	NaN	NaN	2.500000	NaN
25%	NaN	NaN	NaN	NaN	3.100000	NaN
50%	NaN	NaN	NaN	NaN	3.800000	NaN
75%	NaN	NaN	NaN	NaN	4.400000	NaN
max	NaN	NaN	NaN	NaN	5.000000	NaN

	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900	3900	3900	3900.000000	3900	3900
unique	6	2	2	NaN	6	7
top	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	675	2223	2223	NaN	677	584
mean	NaN	NaN	NaN	25.351538	NaN	NaN
std	NaN	NaN	NaN	14.447125	NaN	NaN
min	NaN	NaN	NaN	1.000000	NaN	NaN
25%	NaN	NaN	NaN	13.000000	NaN	NaN
50%	NaN	NaN	NaN	25.000000	NaN	NaN
75%	NaN	NaN	NaN	38.000000	NaN	NaN
max	NaN	NaN	NaN	50.000000	NaN	NaN

Missing Data Handling Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.	Column Standardization Renamed columns to snake case for better readability and documentation.	Feature Engineering - Created age_group column by binning customer ages. - Created purchase_frequency_days column from purchase data.
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Data Consistency Check Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.	Database Integration Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.
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4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue by Gender

Compared total revenue generated by male vs. female customers.

gender	revenue
Female	75191
Male	157890

2. High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88

Total rows: 839

Query complete 00:00:01

3. Top 5 Products by Rating

Found products with the highest average review ratings.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. Data Analysis using SQL (Business Transactions)

4. Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

shipping_type	round
Standard	58.46
Express	60.48

5. Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645.00
No	2847	59.87	170436.00

6. Discount-Dependent Products

Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

4. Data Analysis using SQL (Business Transactions)

7. Customer Segmentation

Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	Number of Customers
Loyal	3116
New	83
Returning	701

8. Top 3 Products per Category

Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions

Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
No	2518
Yes	958

10. Revenue by Age Group

Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



Customer Behavior Dashboard

Filters

Subscription Status

No Yes

Gender

Female Male

Category

Accessories

Clothing

Footwear

Outerwear

Shipping Type

2-Day Shipping

Express

Free Shipping

Next Day Air

Standard

Store Pickup

Key Metrics

3.9K

Number of Customers

\$59.76

Average Purchase Amount

3.75

Average Review Rating

% of Customers by Subscription Status

Yes 27%

No 73%

Revenue by Category

Sales by Category

Revenue by Age Group

Sales by Age Group

Donut chart showing subscription status: Yes (27%) and No (73%)

Bar chart showing revenue by category: Clothing (90K), Accessories (60K), Footwear (30K), Outerwear (15K)

Bar chart showing sales by category: Clothing (1800), Accessories (1200), Footwear (600), Outerwear (300)

Horizontal bar chart showing revenue by age group: Young Adult (55K), Middle-aged (50K), Adult (45K), Senior (40K)

Horizontal bar chart showing sales by age group: Young Adult (1000), Middle-aged (900), Senior (800), Adult (700)

6. Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers.

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control.

Product Positioning

Highlight top-rated and best-selling products in campaigns.

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.